THE JOURNAL OF PHYSICAL CHEMISTRY LETTERS
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Transfer Learning from Simulation to Experimental Data: NMR Chemical Shift Predictions

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Cite This: *J. Phys. Chem. Lett.* 2021, 12, 3662−3668 Read Online

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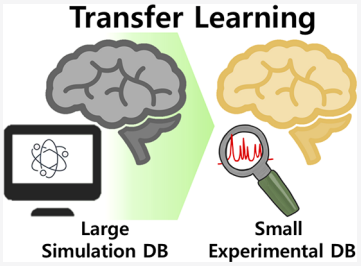
ABSTRACT: An accurate prediction of chemical shifts (δ) to elucidate molecular structures has been a challenging problem. Recently, noble machine learning architectures achieve accurate prediction performance, but the difficulty of building a huge chemical database limits the applicability of machine learning approaches. In this work, we demonstrate that the prior knowledge gained from the simulation database is successfully transferred into the problem of predicting an experimentally measured δ. Although both simulation and experimental databases are vastly different in chemical perspectives, reliable accuracy for δ is achieved by additional training with randomly sampled small numbers of experimental data. Furthermore, the prior knowledge allows us to successfully train the model on the more focused chemical space that the experimental database sparsely covers. The proposed approach, the knowledge transfer from the simulation database, can be utilized to enhance the usability of the local experimental database.



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Elucidating nuclear magnetic resonance (NMR) spectroscopic results is frequently employed with the help of fast and accurate prediction of chemical shifts (δ).[1−8] A quantum mechanical (QM) simulation provides accurate isotropic magnetic shielding constants (σ), which are utilized to estimate δ within a manageable computational cost. Theoretically speaking, σ and δ have the following linear relation

$$\delta = \frac{\sigma_{\mathrm{ref}} - \sigma}{1 - \sigma_{\mathrm{ref}}} \tag{1}$$

where σ_{ref} is the isotropic shielding constants of arbitrary selected samples for each element.[9] Instead of this rigorous transformation, in practice, a simple linear fitting is often used to cancel out potential errors such as the bias of QM calculations and NMR instruments.[7,10]

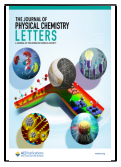
On the other hand, instead of computing σ values based on the quantum theory, direct inferences of δ from accumulated NMR results by a machine learning (ML) model are actively studied.[4−8] In the field of NMR spectrum prediction, the methods to utilize the existing database have been proposed even before the introduction of ML by searching the chemical shifts of atoms that have the same local chemical environment to the target atom.[3,11] The latest ML models reflect the entire chemical environment of molecules so to improve the quality of predictions.[6,8] Despite such successes of ML models, their applicability is still limited by demands of a large database for a model training. Extending the applicable range of ML methods toward the chemical space sparsely covered by existing data becomes a major challenge.[8,12]

In order to overcome the limitation of data availability, the way to utilize the knowledge learned from one problem to solve another problem, named transfer learning, is widely applied in many ML studies.[13,14] In some ML applications such as translation and image recognition, transfer learning has become a standard tool.[15,16] For the chemical problems, several transfer learning examples to predict the high quality of thermodynamic properties of molecules are reported.[5,17,18]

In this report, we describe the construction and training of ML models to predict chemical shifts only with a small amount of experimental data by utilizing transfer learning. We assess the performance of transfer learning approaches on both ^{1}H and ^{13}C spectra according to the size of the experimental database. Also, at the end of the paper, we will also investigate the performance of our approach on a more focused chemical space by training with molecules having a particular chemical environment. It may be helpful to individual research groups who want better performance by combining existing ML models with their local database on the chemistry in which they have interest.

In order to implement transfer learning, we use QM9 and NMRShiftDB2 for the simulation and experimental database, respectively. The QM9 database, which was first published by Ramakrishnan et al., contains 3D structures of small organic molecules and their properties obtained from QM calculations.[19] In addition to the properties provided by the original QM9 database, other properties are also available from its variants.[20−22] Bratholm et al. published results of NMR calculations for molecules belonging to the QM9 database.[23]

Received: February 22, 2021
Accepted: March 31, 2021
Published: April 7, 2021



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https://doi.org/10.1021/acs.jpclett.1c00578
J. Phys. Chem. Lett. 2021, 12, 3662−3668

This database includes magnetic shielding tensor and spin−spin couplings and is freely downloadable from the webpage.[24] From the magnetic shielding tensor, we compute σ by a simple arithmetic operation, one-third of the trace of the magnetic shielding tensor.[9] The experimental database that we use is originally from the open database for NMR spectra, NMRShiftDB2, and further preprocessed by the previous studies.[4,25,26] The released database provides molecular structures in SMILES format and the corresponding chemical shifts for each element to be measured. SMILES structures do not explicitly include 3D molecular structures, so we reconstruct atomic positions using *EmbedMolecule* and *GetConformer* functions implemented in RDKIT (version: 2020.03.6).[27] In order to mitigate the effect of randomness of the structure reconstruction, we perform data augmentation on the experimental databases. We generate five different structures for each of the training molecules of the experimental database.

In order to assess the validity of the obtained 3D structures, the two InChI codes are generated from the SMILES and the corresponding 3D structure through Openbabel (version: 2.4.1), and their identity is confirmed in the same way as the previous report.[28] In total, 66 and 52 molecules fail to construct a 3D geometry from SMILES, and 663 and 435 molecules do not pass the InChI validation for ^{1}H and ^{13}C databases, respectively. A more detailed description of data preprocessing and its results can be found in Table S1.

Table 1 lists summary statistics of both simulation and experimental databases. Since both databases provide training and test subsets split by random selections, we do not perform additional splitting but use the subsets as they are named. Although the sizes of the experimental databases we used are large enough for individual researchers to construct, the simulation database contains more molecules than even the sum of two experimental databases.[3,7,29] Both subsets of the simulation database are fully utilized to learn σ values while only 1, 4, 7, and 10% of randomly sampled molecules from the training subset of the experimental database are used for training of δ values. All molecules belonging to the experimental test subset are used to evaluate test losses.

Table 1. Summary Statistics of Both Simulation and Experimental Databases

database		no. of molecules: training subset	no. of molecules: test subset	no. of heavy atoms per molecules: range	no. of heavy atoms per molecules: mean	constituent elements
expt.	^{1}H	9657	2411	(1, 46)	16.97	H, C, N, O, F, P, S, Cl
	^{13}C	21 116	5304	(1, 40)	14.16	H, C, N, O, F, P, S, Cl
simulation		85 003	45 772	(1, 9)	8.80	H, C, N, O, F

The simulation and experimental databases have different characteristics in aspects of molecular size and constituent elements. The molecules belonging to the simulation database contain one to nine heavy atoms, while ^{1}H and ^{13}C databases consist of molecules having up to 46 and 40 heavy atoms, respectively. Even on average, the experimental databases include larger molecules than that of the simulation database. The detailed distribution of molecules according to a number of heavy atoms can be found in Figure S1. On the other hand, P, S, and Cl appear only in the experimental database, while H, C, N, O, and F elements appear in both databases. The fraction of molecules having P or S or Cl among all molecules in the experimental database can be found in Figure S2(a,(b). Among the three elements, molecules with P are the least distributed (1−5%). The molecules with S and Cl account for more than 10% each.

Among the various ways to implement transfer learning, we employ network-based deep transfer learning to transfer the knowledge obtained from the simulation database to the experimental database.[30] This method does not require any explicit mappings between prior knowledge and the target problem. Network-based deep transfer learning typically consists of two steps: a pretraining and fine-tuning. In the pretraining process, the model parameters are fitted to represent the knowledge for not the target but the related problem. The fine-tuning procedure reoptimizes the pretrained parameters to solve the target problem. The prior knowledge from pretraining prevents overfitting and enhances the performance of a model.[14,31]

We implement our model with message passing neural network (MPNN) layers to preserve the size extensivity, since our model needs to be trained with two databases having different sizes of molecules. There are various variants of the MPNN layer depending on the selection of the update and message functions.[32,33] We employ the edge network and gated recurrent unit for the message and update functions. For initial node and edge features, the embedded atomic numbers and interatomic distances are used. Atomic numbers are embedded into a vector of size 256 by entity embedding. Interatomic distances (d) are transformed into a vector by a Gaussian kernel ($e^{-\alpha(d-d_j)^2}$) embedded with the predetermined d_j values. For d_j, we set an equidistant mesh on [0, 15] with 0.1 Å spacing, and we use 0.1 for α. We stack four MPNN layers to represent interaction among atoms in molecules. The output features of a MPNN layer are used as the input node feature of the next MPNN layers, while the same edge feature from edge embedding is used for all MPNN layers. The outputs of the fourth MPNN layers are mapped into the normalized σ or δ values through the corresponding element's readout layer. For the prediction of δ values, Monte Carlo (MC) dropout is employed as in a previous study, while the conventional dropout layers are employed for σ values.[6,8] MC dropout provides a systematic evaluation of uncertainty, and it is often used to determine the rejected predictions that will not be included in the error evaluation.[6,8,34] The predicted δ and its uncertainty are determined by 15 independent inferences with the same molecular geometries.

For both pretraining and fine-tuning, the L_1 loss function and Adam optimizer are employed, and a batch size is set as 4. In pretraining, a learning rate is initially set as 10^{-4} and is reduced by a factor of 0.5 if the test loss consecutively increases over 5 epochs. In order to avoid an explosive increase of loss in fine-tuning, a learning rate linearly increases up to η_{max} (= 10^{-4}) during the first T_{init} (= 50) epochs. From the T_{init} epoch to the last epoch (T_{max} = 200), the learning rate at T epoch is determined as $\eta_{max}*\cos\left[\pi\frac{T-T_{init}}{T_{max}-T_{init}}\right]$. All aforementioned models are implemented and trained with pytorch (version 1.6.0).

All parameters of pretraining models are randomly initialized. In contrast, except for readout layers and embedding for atomic numbers, the initial parameters of the

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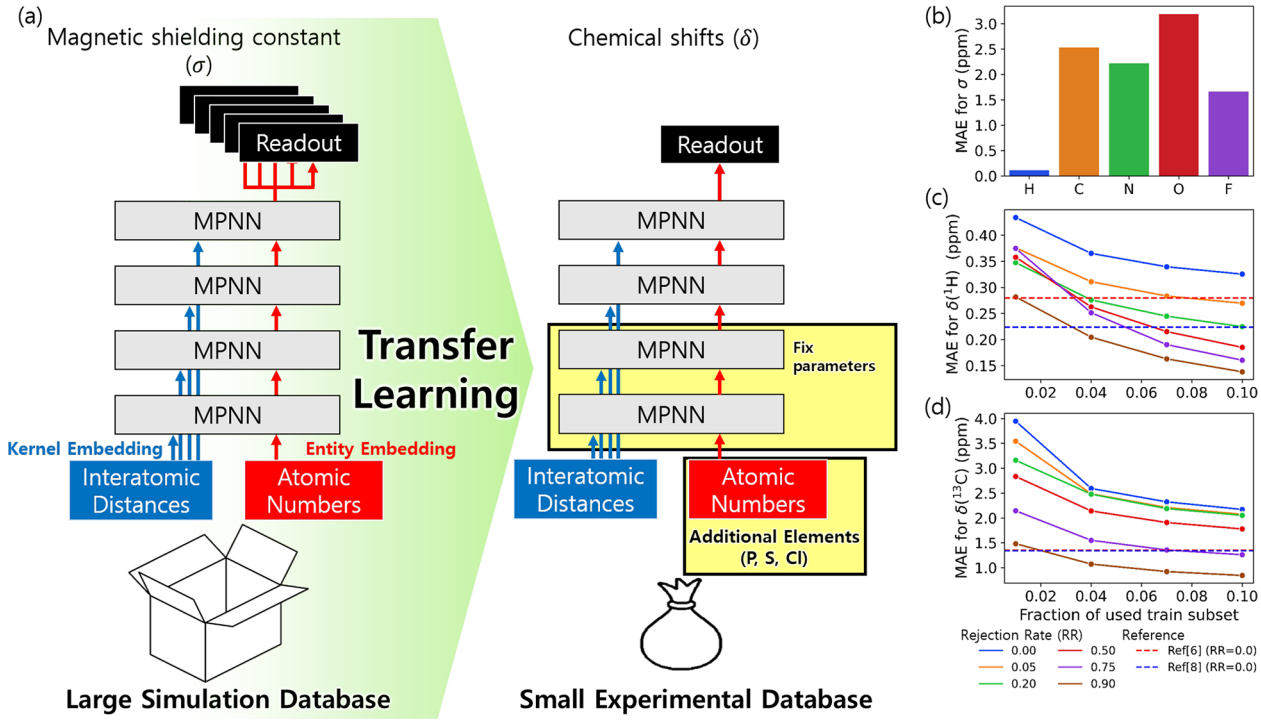

Figure 1. (a) Schematic representation of the model architecture and transfer learning process. (b) Mean absolute errors (MAEs) of a pretrained model on five constituent elements of the simulation database. MAEs of fine-tuned models according to the ratio of data used for fine-tuning the entire experimental train set for (c) ^{1}H and (d) ^{13}C cases.

fine-tuning model are loaded from the pretrained model. Due to the different architecture of the readout layers, the parameters of readout layers for fine-tuning are randomly initialized instead of the pretrained values being loaded. The embedding vectors for P, S, and Cl are not trained during pretraining, so embedding vectors for those elements are randomly generated. (See the right panel of Figure 1(a).) Additional embedding vectors for the atomic number are required, because the experimental database contains elements not included in simulation database. The schematic representation for overall architectures can be found in Figure 1(a). In addition to the modification of architectures, the training process needs to be changed. The parameters of first two MPNN layers are fixed during the fine-tuning to prevent overfitting caused by the small size of training database.

For the pretrained model, the mean absolute error (MAE) of each element and overall data can be found in Figure 1(b). The test MAEs of heavy elements are 1.7−3.2 ppm, while H records 0.1 ppm. This is because the distribution of the isotropic shielding constants of ^{1}H shows a distinctively smaller deviation than other elements. (Figure S3(a,c) shows the distribution of isotropic shielding constants of each element.) For both heavy elements and hydrogen, the measured discrepancies between the pretrained model and QM calculation are less than the discrepancies among QM calculations with different computational conditions (e.g., solvation models, functionals, and basis sets).[2]

Despite successive learning of σ values, an accurate prediction for chemical shifts of the experimental database needs to be further investigated. To assess the transferability of the pretrained model, we evaluate σ for the experimental training molecules and fit those values to the corresponding δ from the experimental databases with a first-degree polynomial. The obtained slope and intercepts are used to derive δ for a test molecule from σ that is the result from the pretrained model. Since the pretrained model is designed to work with only H, C, N, O, and F elements, some experimental database molecules with P, S, and Cl are excluded in performing explicit transformations.

The pretrained model performance on both $\delta(^{1}\mathrm{H})$ and $\delta(^{13}\mathrm{C})$ is worse than its accuracy on σ; the measured MAEs for $\delta(^{1}\mathrm{H})$ and $\delta(^{13}\mathrm{C})$ are 1.0 and 8.6 ppm, respectively. The distributions of the measured errors according to the size of δ for the test subsets are plotted in Figure 2. In both subplots, positive errors for large values and negative errors for small values are recorded. Especially in the range of a large $\delta(^{1}\mathrm{H})$, cases with large positive errors that are highly correlated with δ from the experimental database are observed. Up to our analysis, it is difficult to determine the origin of the correlation between the error of the pretrained model and δ from the experimental database, but an inhomogeneous experimental condition can be one of the possible error sources. The slope and intercept obtained from the least-squares fitting are determined to minimize the error of the majority of data measured at ordinary conditions. Since our simulations do not include various experimental conditions such as temperature and solvent effect, the model trained with the simulation database would not properly consider effects from experimental conditions. The disagreement shown in Figure 2 indicates that the machine learning model trained with simulation databases may not perform well on experimental data even though the model gives sufficiently accurate results for simulation data.

In addition to the accuracy, the explicit transformation is only useful for the molecules that do not include P, S, and Cl elements, because the embedding vectors for those elements are not introduced in pretraining processes. Even though a QM simulation is a powerful tool to elucidate an NMR spectrum,

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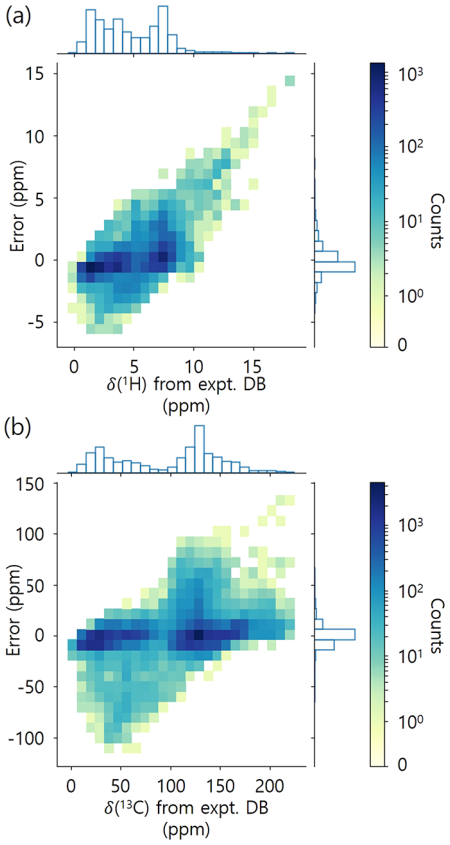

Figure 2. Distribution of prediction error of both (a) $\delta(^{1}H)$ and (b) $\delta(^{13}C)$ that is derived from explicit transformation of isotropic shielding constant inferences.

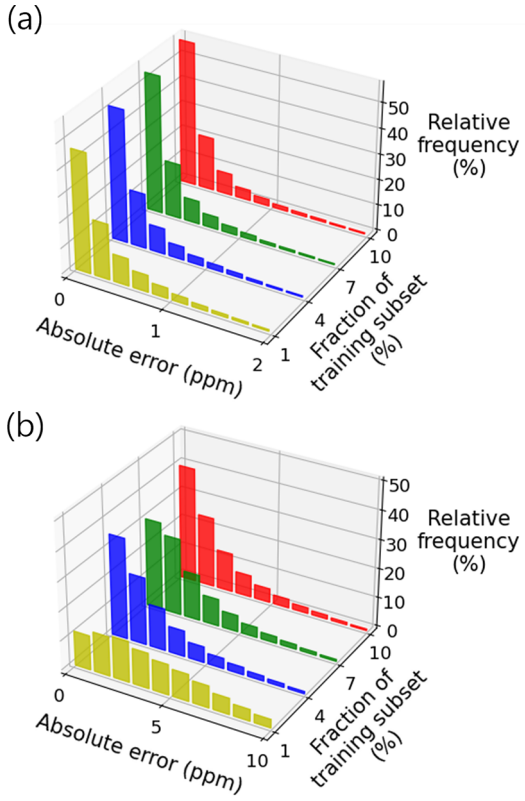

Figure 3. Distribution of absolute error according to the fraction of used training subsets for (a) ^{1}H and (b) ^{13}C databases.

and a state-of-the-art model can learn its results well, learning QM results cannot be an immediate solution for experimentally measured chemical shifts.

We transfer the knowledge gained from the simulation database by additional training with randomly sampled experimental data for accurate predictions of experimental chemical shifts. The MAEs of $\delta(^{1}H)$ and $\delta(^{13}C)$ predicted by the fine-tuned model are plotted in Figure 1(c) and (d), respectively. Both subplots represent mean absolute errors (MAEs) of chemical shifts for the test subset according to the rejection rate (RR) and the fraction of the sampled training subset for the fine-tuning. RR indicates the ratio of discarded prediction values on the error evaluation. As RR increases, a greater ratio of predictions with high uncertainty is rejected. The dashed lines indicate the reference values (RR = 0) from machine learning models training with all training molecules.[4,6]

As we expected, the greater the fraction of data that is used in fine-tuning, the higher the accuracy that is achieved. Also , depending on the choice of RR values, our model gives a smaller MAE than the reference model trained with the whole training subset. The convergence of MAE with respect to the fraction of training subset is also observed in another reference model, as shown in Table S2. However, one significant difference compared to the reference model is sufficient accuracy even in the 1−4% range of the training subset. This is because the interactions among atoms learned from the pretraining are included in our approach.

Figure 3 shows the error distribution of ^{1}H and ^{13}C chemical shifts according to the fraction of sampled training subset. As the sampled fraction increases, the error distribution becomes sharp and close to zero in both cases. However, our results still have a broader distribution than reference results, which use the whole experimental database. (Reference distributions can be found in Figures 2 and 3 of the previous study.[6]) We present Figure S4 to verify whether the correlation between prediction error and δ from the experimental database exists even for a fine-tuned model like the pretrained model. For inferences from the fine-tuned model, the strong correlation observed in Figure 2 is not shown. The fine-tuned model is trained with at least a part of the experimental data so that various effects from experimental measurements are mutually learned during fine-tuning.

In order to figure out the origin of the remaining error, we investigate the effect of the unseen elements in the pretraining. Two different MAEs of molecules having those elements are plotted in Figure 4. The blue bars indicate the MAEs of all prediction values from the molecule having a particular element. The orange bars indicate the MAE of prediction values for neighboring atoms. For the chemical shifts of hydrogen, there are no cases in which a hydrogen atom is connected to Cl or P; therefore, the orange bar is plotted for only the S case. The red dashed lines indicate MAE for overall test molecules. In both ^{1}H and ^{13}C cases, the blue bars are close to the red dashed lines, while the orange bars show much larger values than the red dashed lines. It means that P, S, and Cl atoms lead to small or no increase of errors in average, but the atoms directly bonded to an unseen element cause large errors. A considerable amount of molecules in the experimental database include those elements, but the NMR-active atoms bonded to P, S, and Cl are not sufficiently distributed in our experimental database (see Figure S2(c,d).)

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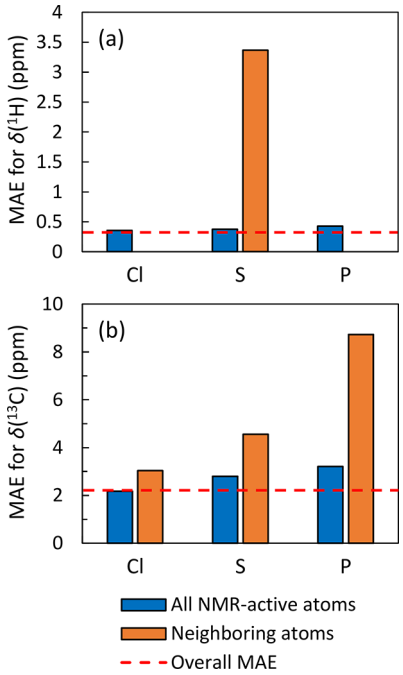

Figure 4. Mean absolute errors (MAEs) on test molecules having the unseen elements in the simulation database for (a) ^{1}H and (b) ^{13}C chemical shifts. The blue and orange bars represent the MAEs of prediction values for all NMR-active atoms and the atoms bonded to the corresponding element, respectively. The red dashed line indicates an MAE of overall test molecules. All errors are evaluated from the inference results of a fine-tuned model with randomly selected 10% of training subsets.

The most straightforward manner to enhance prediction accuracy is to feed more data to a model. However, 10% of the training subset already contains an unaffordable number of molecules by a single research group. A local database that a single research group can create is usually much smaller than our database but densely covers a specific part of the chemical space. Each research group is interested in enhancing the accuracy of inferences on the area that their local database covers instead of building a general-purpose model. Therefore, we demonstrate the concept of transfer learning to enhance prediction performance on a particular chemical space having the largest errors.

In order to investigate the potency of transfer learning on the rare chemical environments, we sample molecules with thiol and phosphate and perform fine-tuning to enhance the accuracy for the sampled molecules. As shown in Figure 4, the randomly sampled subgroups of the experimental database do not sufficiently train our model for a molecule with sulfur and phosphate. We sample all molecules with phosphate from a ^{13}C database and choose a molecule with thiol from ^{1}H, for which the training with a randomly sampled subgroup fails to achieve a high accuracy. The number and fraction of sampled molecules are summarized in Table2. More detailed statistics for subgroups can be found in Table S3.

The average accuracy of fine-tuning with the target sampling and random sampling (10%) on the test subset for both subgroups are plotted in Figure 5. The red and green bars in Figure 5 indicate the MAE of the models trained with random sampling and target sampling, respectively. In both thiol and phosphate subgroups, the target sampling shows almost one-half of MAE values for neighboring atoms compared to the

Table 2. Number of Molecules and Percentage with Respect to Whole Subset for Subgroups

subgroup	training subset	test subset
thiols in ^{1}H database	75 (0.8%)	26 (1.1%)
phosphate in ^{13}C database	358 (1.7%)	80 (1.5%)

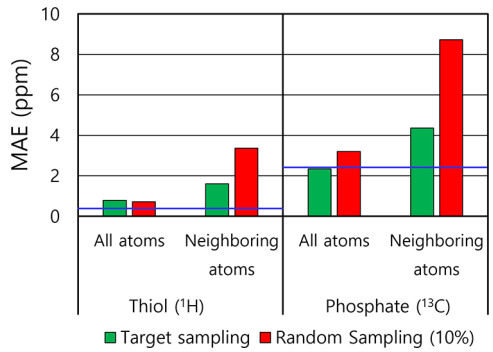

Figure 5. Mean absolute errors (MAEs) of fine-tuned models with randomly selected data (red bars) and target sampled data (green bars). Blue lines represent overall MAEs of the model trained by a randomly sampled training subset. "All atoms" represents MAEs of all NMR-active atoms in the selected molecules, and "Neighboring atoms" represents MAEs of atoms having a bond with the S or P atoms.

random sampling cases, which indicates a huge accuracy enhancement. More precisely speaking, the mitigation of overestimation observed in random sampling results can be found in target sampling. (See Figure S5).

On the other hand, when assigning structure based on the NMR spectrum, the absolute value of the chemical shifts and their relative positions are essential. Hence, accurate predictions for other NMR-active atoms in a molecule are required. For the MAE of all NMR-active atoms ("All atoms" in Figure 5), the measured error of target sampling on the phosphate subgroup is smaller than the error of random sampling on both the phosphate subgroup (red bar) and the overall test subset (blue line). This means that the model trained with the target sampling provides more accurate predictions for at least molecules containing phosphorus, even though the size of target sampling is 7 times smaller than random sampling. On average, target sampling records a larger MAE for all NMR-active atoms than random sampling by 0.06 ppm. This is because the number of training molecules in the thiol subgroup is only 75. Considering the fact that the random sampling for ^{1}H includes ∼1K molecules, the recorded performance degradation is acceptable.

Figure 6 shows molecular structures and chemical shift values of molecules whose prediction errors are most reduced in average by the target sampling. Although machine learning models provide chemical shift values for all atoms of target elements, the δ values are notated only for atoms for which the experimental values are available. As shown in Figure 5, the model trained with target sampling shows overwhelming accuracy for the neighboring atoms. Overall prediction values of models trained with thiol and phosphate subgroups can be seen in Figure S5, which also represent a great performance enhancement for neighboring atoms while keeping accuracy on the rest of atoms

In summary, reliable chemical shift predictions for both ^{1}H and ^{13}C are achieved by transfer learning with a large

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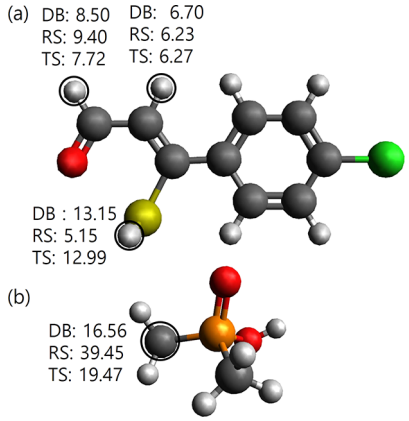

Figure 6. Structures and chemical shift values of test molecules with the largest increase of predictive performance through target sampling among test molecules of (a) thiol and (b) phosphate subgroups. "DB", "RS", and "TS" indicate δ values from the experimental database, random sampling with 10% of a training subset and target sampling, respectively. (Units: ppm).

simulation database and small experimental databases. Also, we apply transfer learning on the more focused chemical space and obtain a more accurate machine learning model. By fine-tuning with a tiny set of molecules having thiol (75 molecules) and phosphate (358 molecules), accurate predictions for not only atoms neighboring the sulfur and phosphate but also all NMR-active atoms in molecules are achieved.

Constructing a more extensive database with more qualified data can be an ultimate solution for NMR prediction via machine learning modeling. However, it is not always feasible, especially for individual research groups. Although the model trained with only simulation data fails to extend its predictability on the experimental database, the knowledge gained from the simulation database allows the model can be sufficiently trained with a small number of experimental data. Here, we demonstrate this transfer learning for chemical shift predictions, but our approach is not limited to a particular molecular property. Our approach potentially extends the applicability of in-house experimental databases that remain as chemists' experience. Thanks to the tremendous advance in computing performance, the price for constructing massive simulation databases becomes lowered and accessibility of these databases becomes improved. Hence, transfer learning from simulation to experimental data, demonstrated here, will be a competitive way of collaboration between theorists and experimentalists.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpclett.1c00578.

Detailed description of preprocessing of a simulation database; statistics of experimental and simulation databases; performance of machine learning model with random sampling; details of subgroups and target learning results (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors thank Lars A. Bratholm (University of Bristol, UK) for kind discussion on the simulation database. This work has been carried out with the support of the National Research Foundation of Korea [NRF-2019M3E5D4065P6812]. The computing resources are supported by KISTI[KSC-2020-CRE-0117] and UNIST.

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https://doi.org/10.1021/acs.jpclett.1c00578
J. Phys. Chem. Lett. 2021, 12, 3662−3668